

HW 5

- **Mirror plane (m)**

Transformation matrix for m including x, y axes

$$\begin{pmatrix} x \\ y \\ -z \end{pmatrix} = \Delta \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \Delta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Exercise I:

Please give the transformation matrix when m include x, z axis ?

When m include y, z axis ?

Exercise:

Please derive all the transformation matrixes for L^6 .

Exercise for Theorem 1:

$$L^6 \times L^2_{\perp} \rightarrow ?$$

Exercise for converse theorem :

When L^2 intersects with L^2 with 15° , which symmetry axis can be generated?

Please give the symmetrical patterns to demonstrate the combination of symmetry elements above.

Exercise of theorem 3:

$$L^3 \times P_{//} \rightarrow ?$$

When two P intersect with 60° , which symmetry axis will be generated?

Please give the symmetrical patterns to demonstrate the combination of symmetry elements above.

Exercise: to build up a paper model of the above polyhedron.

Please post your photos here!

1. tetragonal tetrahedron

2. Zongzi

3. ditrigonal scalenohedron

Please prove transformation matrix of L^n

$$\begin{pmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Please prove transformation matrix of L_i^n

$$\begin{pmatrix} -\cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & -\cos \alpha & 0 \\ 0 & 0 & -1 \end{pmatrix}$$